
Narrowing Without Converging: How Interactive LLM Agents Search Under Feedback

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Abstract

1 Interactive agents should use feedback to converge on correct answers, yet how
2 they search remains poorly understood. We study three Qwen model tiers on 22
3 Turtle Soup puzzles using scored answer checkpoints and embedded trajectories
4 of successive questions. Accuracy improves for only the first 10 of 30 rounds,
5 while semantic stride contracts four- to fivefold. Longer budgets, Oracle quality,
6 and puzzle under-determination do not explain this plateau. Feedback also redi-
7 rects search poorly: after refutations, all three tiers move toward regions already
8 ruled out. Agents can also reach solution-grade stories and subsequently aban-
9 don them. These results identify two distinct failures, ineffective feedback use
10 and failure to preserve strong hypotheses, that endpoint accuracy cannot distin-
11 guish. We release the evaluation harness, process measures, verified puzzle set,
12 and per-game logs.

13 1 Introduction

14 Consider a Turtle Soup puzzle: *A man’s body lies in the desert, clutching half a matchstick; luggage*
15 *and clothing are scattered around him.* The solver sees only this surface and must recover the hidden
16 explanation through yes/no questions. In the solution, a hot-air balloon was losing altitude. After
17 discarding their luggage, the passengers drew lots, and the man holding the short matchstick was
18 thrown overboard.

19 Existing evaluations can determine whether an agent eventually recovers this story, but not how it
20 searches. The same endpoint score may describe an agent that never leaves its initial hypothesis
21 and one that discovers the answer but later abandons it. Existing Turtle Soup benchmarks empha-
22 size outcomes rather than search dynamics (Appendix A) [1, 2, 3]. This leaves a central question
23 unresolved: how does feedback change an agent’s next move?

24 Turtle Soup provides a controlled setting for answering this question. The interaction is textual, the
25 Oracle supplies a discrete yes/no/irrelevant signal, and each puzzle is deliberately under-determined.
26 We introduce two process-level probes. After every Oracle response, the Questioner produces a
27 complete checkpoint reconstruction, allowing accuracy to be tracked across rounds. We also embed
28 each question to trace the association trajectory and measure its per-round semantic stride. We
29 evaluate three Qwen model tiers on 22 human-played puzzles.

30 We test three hypotheses. **H1**: refutations progressively constrain the hypothesis space, so shorter
31 strides should accompany convergence. **H2**: feedback redirects the next question, with refutations
32 pushing the search away and confirmations promoting local refinement. **H3**: once an agent reaches
33 a solution-grade reconstruction, it preserves and commits that answer.

34 Our analysis shows that interactive search can narrow without converging. It also separates two fail-
35 ures hidden by endpoint accuracy: circling around an early hypothesis and abandoning a stronger

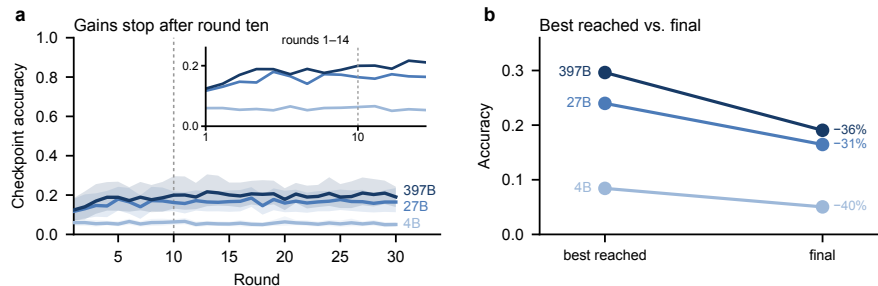


Figure 1: (a) Accuracy of the answer each tier would give at each round (bands, 95% cluster-bootstrap CIs over 22 puzzles); dashed rule at round 10; inset, rounds 1–14. (b) The best answer each game reached, against the answer it finally stands on.

earlier reconstruction. Finally, we establish an important measurement boundary: embedding geometry can describe these trajectories but cannot reliably score their success. We release the evaluation harness, process instruments, verified puzzle set, and per-game logs.

2 Framework and experiments

Task and harness. A **Questioner** sees only the puzzle’s visible *surface* and asks one yes/no question per round, following the SPLAT player–judge protocol [2]. An **Oracle** holds the hidden story and answers yes, no, or irrelevant. The Questioner may submit a final story at any round and must do so when the round budget is reached. The open-source harness records every exchange and the seed for each game.

Checkpoint and commitment instrumentation. In E1, the Questioner writes its best complete reconstruction after every Oracle response. We score each checkpoint, so every point on the accuracy curve represents a full answer based on the evidence available at that round. We separately record *commitment*, defined by whether and when the Questioner voluntarily submits a final story. Together, these measures convert an endpoint benchmark into a scored process.

Puzzles. To ensure that the intended explanations are demonstrably recoverable, we restrict evaluation to puzzles with a public record of human play. The resulting set contains 22 puzzles from a Chinese Turtle Soup community site, each retaining its source URL. Surfaces contain 9–108 characters, solutions contain 13–200 characters, and the set includes 97 annotated key clues. The repository separates this verified family from a quarantined generated family, and automated tests prevent the two from being mixed.

Puzzle under-determination. Each puzzle receives an **under-determination** score that measures how far the surface alone leaves a solver from the hidden solution. We generate twelve complete stories from the surface without Oracle feedback and identify the guess closest to the solution in embedding space. Under-determination equals one minus this maximum similarity, with higher values indicating a larger initial gap. We use the score as a puzzle-level covariate; Appendix B provides its distribution and validation.

Outcome scoring. The outcome score combines **clue recall** (70 points) and **causal-logic identity** (30 points), as detailed in Appendix C. The logic component recognizes causally equivalent answers expressed through different wording. The clue component anchors answers to the annotated evidence and prevents semantically distant alternatives from receiving high scores.

Association-trajectory instrumentation. Our second instrument analyzes the questions themselves, adapting forward flow [7] and the Divergent Association Task [8] from human psychometrics. For each round, we extract keywords with jieba, embed them using bge-small-zh-v1.5, and average the normalized embeddings into a unit vector q_t . All distances are cosine distances.

We use two anchors. The surface-only anchor represents terms from the visible puzzle surface, whereas the solution-aware anchor combines surface, solution, and clue terms. The primary measures are the **stride**, $s_t = d(q_t, q_{t-1})$, and the **anchor distance**, $h_t = d(q_t, \mathcal{A})$, computed separately for each anchor. H1 predicts that contraction in stride should accompany movement toward the solution and improved checkpoint accuracy. H2 is tested by pairing each Oracle response at round t with the next stride s_{t+1} and by comparing movement toward previously confirmed and refuted regions. H3 is tested by comparing solution-grade checkpoint peaks with subsequent commitment and final answers. The lexical `question_novelty` metric serves as an ablation that cannot capture movement expressed through synonyms.

Experimental designs. E1 plays each game for up to 30 rounds and scores a checkpoint after every round. E2 imposes hard caps of $\{5, 10, 15, 20, 25, 30\}$ rounds and scores only the forced final answer, providing a no-checkpoint control. E3 analyzes the E1 trajectories using $\{s_t\}$, $\{h_t\}$, response-conditioned movement, and checkpoint commitment. The keyword extractor is fixed across models, preventing any model from being evaluated through its own vocabulary.

Model grid and validity controls. We evaluate Qwen3.5-4B, Qwen3.6-27B, and Qwen3.5-397B-A17B on 22 puzzles with three seeds. E1 therefore contains 198 thirty-round games, while E2 contains 1,188 games across the six caps. DeepSeek-V3.1 serves as both Oracle and causal-logic judge. It is cross-family with every Questioner and achieves 95% accuracy on held-out yes/no probes. Every model call uses `max_tokens=2048`, disabled thinking, temperature 0.2, and a 50,000-token per-game budget. Analyses isolate the documented token-exhaustion artifact affecting the smallest tier; no harness failures occurred. Because the Questioners differ in training generation and architecture, we compare model tiers rather than claim a scaling law.

3 Results

Interaction improves accuracy only during the first ten rounds. Checkpoint accuracy increased through approximately round 10 and then plateaued at every tier (Figure 1a). From rounds 10 to 30, accuracy changed by -0.009 , $+0.003$, and -0.013 for 4B, 27B, and 397B, respectively; all confidence intervals included zero. Each tier also ended below its best checkpoint accuracy (Figure 1b).

The plateau was not explained by the round budget. Final accuracy was insensitive to the cap for the two larger tiers. The apparent decline at 4B coincided with token exhaustion, which rose from 0/66 to 63/66 games across caps; no decline appeared among non-exhausted games. The Oracle achieved 95% accuracy on held-out probes, and 68–74% of its responses were “no”, ruling out a collapsed feedback channel. The rank correlations between under-determination and round-1 accuracy were $-0.34/-0.54/-0.47$. No relationship was detected with improvement after round 10, any trajectory measure, either score component, or the between-tier gap (all n.s. at $n = 22$; Appendix C).

H1: the search narrows toward the answer. Mean stride decreased monotonically at every tier, falling four- to fivefold from 0.145/0.232/0.182 at round 1 to 0.025/0.053/0.045 at round 29. This contraction did not accompany convergence: accuracy remained flat, and within-tier stride did not predict final performance ($|\rho| \leq 0.16$). The 4B tier had the smallest late-game stride, 0.030 [0.021, 0.039], compared with 0.082 [0.065, 0.098] for 27B, yet never reached a solution-grade answer. Its trajectory therefore contracted around a persistent hypothesis rather than toward the solution. Excluding its 17% truncated or non-interrogative questions changed the mean and late stride only from 0.049 and 0.030 to 0.056 and 0.033, respectively.

H2: the answer redirects the next question. We paired each Oracle response with the next stride and controlled for round position within puzzle and round-bin cells. Let Δs denote stride after confirmation minus stride after refutation. No difference was detected for 4B ($+0.001 [-0.009, +0.011]$), whereas 27B and 397B moved farther after confirmation ($+0.028 [+0.010, +0.046]$ and $+0.009 [+0.002, +0.017]$; Figure 2a). Response-conditioned movement was therefore absent at the smallest tier and opposite to the hypothesized direction at the larger tiers.

Because an informative confirmation could reasonably produce a larger step, we also used a direction-based test that assumed no optimal stride. After a refutation, the next question moved

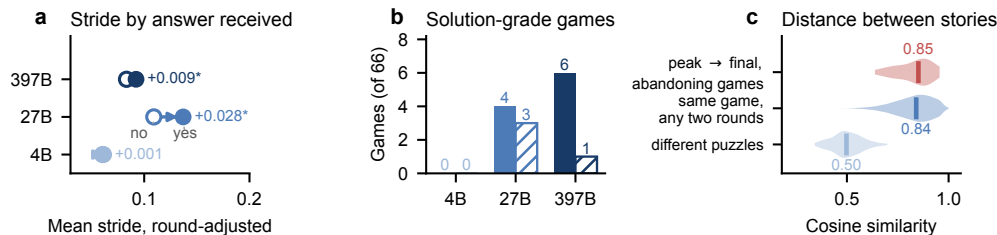


Figure 2: (a) Stride into the next question after refutation and confirmation, adjusted for round position. (b) Games reaching a solution-grade checkpoint and the subset voluntarily committed. (c) Similarity between final and peak stories in abandoning games relative to ordinary within-game variation.

toward regions the Oracle had already ruled out. This effect appeared at every tier: -0.089 $[-0.112, -0.068]$ for 4B, -0.051 $[-0.075, -0.032]$ for 27B, and -0.032 $[-0.047, -0.018]$ for 397B. No corresponding asymmetry appeared after confirmation. Feedback therefore failed to steer any tier consistently away from rejected regions.

H3: the agent stops when it finds the answer. We defined a sustained checkpoint peak of at least 0.5 as *solution-grade*. The 4B tier reached no solution-grade checkpoint. The 27B tier reached solution-grade checkpoints in four games and voluntarily committed in three; 397B did so in six games but committed in only one (Figure 2b). Commitment was rare and late overall, occurring in 11/66, 9/66, and 2/66 games at mean rounds 11.1, 14.2, and 23.5.

The dominant failure differed by tier. The 4B tier circled around an early hypothesis in 39/66 games and abandoned a stronger answer in one. The 397B tier circled in 6/66 games and abandoned a stronger answer in 12. The 397B tier therefore reached useful hypotheses more often but did not reliably preserve or commit them.

Embedding geometry describes abandonment but does not score it. In abandoning games, the median cosine similarity between the final and peak stories was 0.851. This was comparable to the similarity between checkpoints from the same game (0.841) and much greater than that between stories from different puzzles (0.498; Figure 2c). Despite this semantic proximity, abandoning the peak reduced the score by 0.30 on average. The decisive substitution was therefore smaller than the encoder’s resolution.

Consistent with this limit, no systematic drift was detected, and all within-tier coefficients linking stride or drift to best accuracy were bounded within ± 0.035 per standard deviation (Appendix C). Lexical novelty detected only the 4B tier’s repeated questions. The trajectory geometry characterized how agents failed, but it did not predict how well they scored.

4 Limitations and Conclusion

Limitations. Our evidence comes from one game, one Oracle protocol, and three Qwen-family Questioners. The trajectory measures capture how far each question moves in embedding space, not what operation it performs; round position is controlled only by binning, the anchors are not human-derived, and a single encoder may miss small semantic changes that alter correctness. Token exhaustion can mimic inability to form questions; the Oracle’s 95% audited accuracy implies roughly one wrong response per game; and using the same DeepSeek model as Oracle and logic judge may correlate their errors. Under-determination may partly measure puzzle familiarity, though no claim rests on it. The puzzles include death and other dark themes and carry content flags.

Conclusion. Here, the agents narrow without converging: semantic stride decreases four- to fivefold while accuracy plateaus, and feedback fails to steer questions consistently away from rejected regions. The smallest tier mostly circles early hypotheses, whereas the largest tier sometimes reaches solution-grade reconstructions but fails to preserve or commit them. The two failures require different fixes yet appear identical under endpoint accuracy; process-level measurement reveals how search fails even when it cannot predict performance. Whether these patterns extend to other tasks, model families, and protocols remains open.

Declaration of LLM usage

LLMs are the object of study: the Questioner agents, the Oracle, and the causal-logic judge are all LLMs, specified in §3 and Appendix C, and every reported number derives from their logged interactions. Separately, LLM assistants were used to help write analysis code and edit prose; the authors reviewed all analyses, claims, and text, and take full responsibility for them.

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186 A Related work

187 Existing Turtle Soup benchmarks score outcomes. TurtleBench [1], SPLAT [2] (whose interactive
 188 player-judge protocol we adopt), and TurtleSoup-Bench [3] evaluate verification accuracy, solve
 189 efficiency, or faithfulness to the intended story. TurtleSoup-Bench also probes what agents ask next,
 190 but it grades the question against the intended story; none of the three measures the geometry of
 191 the search itself, which is the gap this paper addresses. MUTATE [4] shows that frontier agents
 192 find far fewer mechanism-distinct solution paths than humans; its diverge-then-narrow agent is a
 193 natural intervention to test with our instruments. The trajectory instrument adapts forward flow [7]
 194 and the Divergent Association Task [8] from human psychometrics, and the Small World of Words
 195 norms [9] are the natural source for the human-derived anchor that ours is not yet (§2). Catalogues
 196 of judge bias [5, 6] motivated confining the model judge to the one subscore that objective matching
 197 cannot reach (Appendix C).

198 B The puzzle set

199 **Selection.** The 22 puzzles were hand-picked from the classic repertoire of a Chinese Turtle Soup
 200 community, restricted to puzzles with a public record of human play (§2): a puzzle nobody has
 201 solved carries no evidence that its clues suffice or that the intended leap is reachable. Every candidate
 202 was read and kept or rejected by hand. We rejected puzzles that turn on deduction from stated
 203 evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source.
 204 Surfaces run 9–108 characters (median 34) and solutions 13–200 characters (median 102).

205 **Distribution of under-determination.** Under-determination is defined in §2. Its distribution over
 206 the 22 puzzles is:

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

207
 208 The median is 0.347, the quartiles are 0.273 and 0.410, and the full range is 0.173–0.577: single-
 209 peaked with a thin hard tail. The easiest puzzles are the story the genre is named for (0.17) and the
 210 desert matchstick (0.19); the hardest is a fourteen-character surface reading *our heights differ / a*
 211 *flowerpot broke / our heights are the same* (0.58).

212 **Validity as a covariate.** The score describes the puzzle, not any agent’s performance on it, and it
 213 is independent of rounds used; a difficulty defined as “how many rounds this takes” would make an
 214 accuracy-versus-rounds plot confirm itself. It is not a proxy for surface length ($r = -0.22$, n.s. at
 215 $n = 22$, a weak test). It is also one measure rather than two: a surface loose enough to admit many
 216 mechanisms is, for the same reason, one whose true mechanism takes more reframing to reach, so
 217 *how far* and *how many ways* are not separable here.

218 **Familiarity caveat.** The two puzzles scoring easiest are among the most widely circulated, and
 219 the models’ first-round guesses are consistent with familiarity contributing: on the best-known of
 220 them, the two larger tiers open at 0.61 and 0.53 before any evidence arrives, against 0.24 and 0.36
 221 on a puzzle of almost identical under-determination. On the easy end the measure may therefore
 222 read familiarity as much as difficulty. Separating the two would take a memorisation probe, which
 223 we have not run; no claim in the paper rests on the measure.

224 C Measuring agent performance

225 **Outcome score.** The outcome score combines **clue recall (70 points)** and **causal-logic identity**
 226 **(30 points)** (§2). Clue recall matches the answer against per-puzzle annotated key clues and is
 227 objective and reproducible; matching tolerates paraphrase through character-bigram recall, which

sets a floor on how short an annotated clue may be (below the floor only a verbatim match counts, so all clues are annotated above it). The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is instructed to ignore wording. Because single ratings are unstable on borderline answers, each rating is sampled twice at temperature 0.2 and averaged; the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average. The gap between the two subscores is itself informative: right story, wrong vocabulary.

The logic half earns its thirty points. Every model earns proportionally far more of the judged causal-logic half than of the clue-recall half (397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, 4B 2.9 against 0.2), and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each): models reach the right causal shape well before, and often without, landing the annotated clue content. The 70/30 weighting is a design choice; every conclusion in §3 is readable from the two halves separately, though the absolute plateau level (~ 0.2) depends on the weights.

Feedback response: definition, confound, and both estimates. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; “irrelevant” rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for a “yes” against 15.6 / 15.6 / 15.9 for a “no”) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]). The gap between the two is the point.

The semantic scale of abandoning. Abandoning games are those with a sustained peak ≥ 0.3 whose final checkpoint falls below half of it ($n = 34$). The final story sits at median cosine similarity 0.851 from the peak story, while any two checkpoints of the same game sit at 0.841 (median over all 77,906 within-game pairs from 60 games) and two stories from different puzzles at 0.498 (231 pairs). Across the same peak-to-final pairs, the median score falls from 0.447 to 0.099.

The round-1 prior. The round-1 checkpoint precedes the first Oracle answer and is therefore a cold guess. It holds 49% / 33% / 26% (median, by tier) of the game’s eventual best score, and correlates with it at $\rho = +0.37/ +0.57/ +0.47$ (permutation $p = 0.09/0.003/0.03$, $n = 22$ puzzles with seeds averaged). Under-determination, itself measured from twelve cold guesses by a model cross-family with every Questioner, predicts round-1 quality at $\rho = -0.34/ -0.54/ -0.47$ ($p = 0.12/0.010/0.029$) but not the final score.

Behavioural indicators against puzzle structure. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, drift (both anchors), mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is negative at all three (−0.32/ −0.25/ −0.43) but pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

Retention estimator and permutation null. Retention is the mean per-game final-to-sustained-peak ratio over games with a nonzero sustained peak; the final is the last played checkpoint, and excluding early-committing games changes each tier’s value by ≤ 0.03 (bootstrap 95% CIs $\pm \approx 0.08$). For the null: unit, the game. For each of 2,000 permutations, every game’s checkpoint sequence is shuffled within the game (knowledge-constant, jitter-only null), the sustained peak and the final (last element) are recomputed on the shuffled sequence, and retention is the mean over that tier’s games with a nonzero permuted peak. The reported p is one-sided per tier, the fraction of permutations whose null retention is at or below the observed value. Observed retention (0.69–0.76 by tier) does not fall significantly below this null (0.79–0.84, $p \geq 0.09$), so the peak-to-final gap visible in Figure 1b is reported as consistent with scoring jitter and no claim in the body rests on its size; the commitment record of §3 is behavioural and unaffected by jitter.

Mixed-model coefficients. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept, tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride

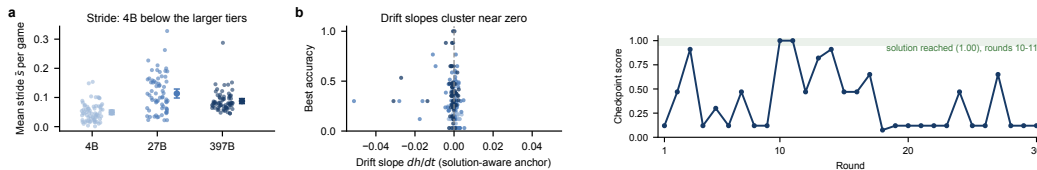


Figure 3: Per-trace geometry ($n = 198$). (a) Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. (b) Drift slope against best accuracy, clustered near zero at every tier.

Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Table 1 shows seven of the thirty rounds.

279 $\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, $p = 0.42$; drift $\beta = -0.011$, $[-0.035, 0.014]$,
 280 $p = 0.39$; stride $\times$ tier $p = 0.20$; surface-anchor drift $\beta = -0.020$, $[-0.044, 0.003]$, $p = 0.097$.
 281 Figure 3 shows the underlying per-trace geometry.

282 **Oracle audit and game-budget outcome.** Model-call settings are given in §2; exhausting
 283 the 50,000-token game budget forces a final answer and is recorded as the game outcome
 284 token_budget. The Oracle audit probes a candidate with 25 hand-crafted (question, expected-
 285 answer) pairs over three puzzles, 21 with a definite yes/no answer and 4 expecting “irrelevant”; the
 286 pass bar is 90% on the yes/no items. The deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no
 287 and 3/4 exact on “irrelevant”. Pilot candidates: a 4B Oracle with thinking disabled, 30% (rejected;
 288 in that configuration, raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score un-
 289 changed); the same model with thinking enabled, 70% (rejected); a same-family flash model, 90%
 290 (excluded for sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

291 **Trajectory and difficulty pipeline.** The embedding pipeline is specified in §2; two details matter
 292 for reproduction. Keyword extraction strips Latin characters and digits before jieba segmentation,
 293 and the round vector is the re-normalized mean of the keyword embeddings. Late-game stride de-
 294 notes the second half of a game’s strides. The under-determination measure prompts GLM-5.3-flash
 295 (cross-family with all Questioners) for the twelve cold guesses and scores closeness as embedding
 296 similarity between guess and solution under the same encoder, not with the composite score, so the
 297 covariate is not coupled to the outcome apparatus.

298 **Artifacts.** Harness, puzzle set with per-item source records, per-game logs (including every check-
 299 point story), and the analysis scripts behind each reported number are released at a public repository,
 300 withheld here for double-blind review and available to reviewers on request through the programme
 301 chairs. Puzzle surfaces and solutions originate from a public community repertoire and are redis-
 302 tributed with source attribution and a content flag.

303 **One game, worked.** Figure 4 and Table 1 show puzzle refsoup_021 (surface: a stormy night,
 304 a run-down temple, two photographs on the wall that seem to stare; by morning the narrator sees
 305 only two windows; solution: “there were never photos on the wall; people were pressing against the
 306 windows, looking in”), played by Qwen3.5-397B, seed 2. Questions and stories are our translations
 307 of the logged Chinese, with stories abridged to the clause carrying the hypothesis. Rounds 22–30,
 308 eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the narrator
 309 woke floating upside down; round 28, whether rising water reached eye level), are in the released
 310 logs.

311 The table shows three things that no mean curve can. **The checkpoint mechanism surfaces knowl-**
 312 **edge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from
 313 earlier answers on a round whose own question was answered “no”. Yet the model never volunteers
 314 a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19
 315 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Circling**
 316 **in fresh words happens at the largest tier too.** Rounds 22–30 are eight lexically distinct re-asks
 317 of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s round-to-**
 318 **round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	Are the ones in the photos people?	yes	The two “photos” are two people nailed or sealed into wall niches.	0.12 (0+12)
3	On waking the next day, is he still in the temple?	yes	The “two photos” are really two windows, mistaken for photos in the dim light.	0.91 (70+21)
4	Are the “two windows” the same objects as last night’s “photos”?	yes	The “photos” were hollowed-out window openings; the fear came from darkness and illusion.	0.12 (0+12)
10	Did frame glass break and fall, leaving open holes?	no	They were never photos, but faces pressed against the outside of the windows.	1.00 (70+30)
11	Were the photos removed, revealing windows behind?	no	The staring figures were people outside the windows.	1.00 (70+30)
19	Were real people hiding behind the photos?	no	Window openings sealed by painted cloth covers; the storm tore them off.	0.12 (0+12)
30	Did rising water lift the narrator higher?	no	Papered-over window openings; the storm stripped the paper away.	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Questions and stories are our translations; the original transcripts are in the released logs.

319 3 → 4) swings the clue half from 70 to 0. The mean curves of §3 average this out, which is why
320 single-game curves are not the unit of analysis.